

Acupuncture Channels & Points I

AC300 / AC375 · VUIM Summer 2026 · Dr. Vivian Zhang, Ph.D.

Print Edition

WEEK

9

Points & Final Exam Review

Five Shu Points · Confluent Points · 15 Collaterals · 12 Cutaneous Regions

Comprehensive Final Exam Master Review · Clinical Evidence Appendix

This Week Covers:

- Acupoint categories: Five Shu (Transport) Points -- full 60-point master table across all 12 meridians
- Eight Confluent Points -- detailed cards, master-couple pairing, location + function
- 15 Collaterals master table + the 3 extra (non-organ) collaterals
- Middle Circuit completion: PC/SJ/GB/LR Divergent Channels + Muscle Regions (confirms Week 8 self-study flags)
- 12 Cutaneous Regions -- definition, functions, all 6 groups with lecture figures
- Final Exam Master Review -- circulation rules, qi-flow directions, and the full 12-meridian summary table
- Clinical Evidence appendix: PCOS/PMOS protocol, urinary incontinence electroacupuncture (JAMA), EMMA robotic massage pilot data

NEXT WEEK: Comprehensive Final Exam (material from Weeks 1-9). No new quiz/homework this week -- see flag inside.

No new CAM or MOA reading pages are assigned for Week 9 in the slide deck or transcript. Dr. Zhang's own instruction is to "review according to the slide I prepared" -- i.e., the Week 9 deck itself, plus cumulative review of all prior weeks' lecture notes and reading assignments, is the assigned final-exam preparation material.

This Week's Framework: Acupuncture Points

General Functions & Categories

Acupoints are specific sites on the human body where the Qi and Blood of the Zang-Fu organs and meridians are transported and infused. They also serve as reflection points for diseases and stimulation points for treatment.

Classified by name, location, and meridian: Fourteen Meridian Point · Extra Point · Ouch (Ashi) Point

Physiology: Infusion of Qi and Blood, nourishing the entire body (Physiology).

Pathology: Clearing the Meridians and Harmonizing Yin and Yang (Pathology).

This Week's 3 Special-Point Categories:

15 Collaterals

Where a Luo-vessel branches off to its paired meridian.

Eight Confluent Points

Connect the 8 Extraordinary Vessels to the 12 Regular Meridians.

Five Shu (Transport) Points

5 points per primary meridian, distal to elbow/knee -- 60 points total.

CONFIRMED THIS WEEK (resolves a Week 8 flag):

CONFIRMED this week: Week 8's materials flagged the GB/LR collateral + divergent-channel narration and all 12 Muscle Region / Cutaneous Region slides as "self-study, not reached live" (Dr. Zhang said she would cover the middle circuit "next week"). The Week 9 deck's own agenda (slide 3) and the Week 9 live transcript confirm this content WAS delivered live this week -- GB/LR collaterals, GB/LR divergent channels, and all 12 Muscle + 12 Cutaneous Regions are no longer self-study. Full diagrams for GB/LR muscle regions are included below; see the Week 8 Study Guide for the complete diagram set covering the anterior and posterior circuits.

NEXT WEEK: FINAL EXAM

CONFIRMED via both the live transcript and the 2026 slide deck's own final slide: NEXT WEEK IS THE COMPREHENSIVE FINAL EXAM (material from Weeks 1-9, 30 questions per Dr. Zhang's verbal answer in class). No new quiz or homework was assigned this week -- Dr. Zhang stated in lecture that Homework 5 (from Week 7/8) was still outstanding for some students and reminded the class to submit it before the final, since grades close after the exam. This Quiz Kit and Cram Sheet are self-test review material for that comprehensive final, not a stand-in for a new graded quiz.

No new CAM or MOA reading pages are assigned for Week 9 in the slide deck or transcript. Dr. Zhang's own instruction is to "review according to the slide I prepared" -- i.e., the Week 9 deck itself, plus cumulative review of all prior weeks' lecture notes and reading assignments, is the assigned final-exam preparation material.

The Five Shu (Transport) Points

60 points -- 5 per meridian x 12 meridians

The term ‘Five Shu Points’ (Wu Shu Xue) collectively refers to a specific group of five acupoints on each of the twelve main Meridians, situated distal to the elbows and knees. They are individually named Jing-Well (Jing), Ying-Spring (Ying), Shu-Stream (Shu), Jing-River (Jing), and He-Sea (He).

CATEGORY	MEANING	LOCATION	CLINICAL APPLICATION
Jing-Well	“Where it emerges is the Jing-Well”	Tips of fingers and toes	First aid, clearing heat, consciousness disorders
Ying-Spring	“Where it flows is the Ying-Spring”	Before the MCP/MTP joints	Feverish diseases, heat-related disorders
Shu-Stream	“Where it pours is the Shu-Stream”	After the MCP/MTP joints	Heaviness in the body, pain/stiffness in joints
Jing-River	“Where it travels is the Jing-River”	Forearms or lower legs	Externally contracted diseases (colds, flu), meridian regulation
He-Sea	“Where it enters is the He-Sea”	Near the elbow or knee joints	Disorders of the six Fu (hollow) organs

Classic of Difficult Questions, Ch. 68: “The well points govern fullness below the heart, the spring points govern fever, the stream points govern heaviness of the body and joint pain, the river points govern asthma, cough, chills, and fever, and the sea points govern rebellious qi and diarrhea.”

Five Shu Master Table — All 12 Meridians

MERIDIAN	JING-WELL	YING-SPRING	SHU-STREAM	JING-RIVER	HE-SEA
ANTERIOR (OUTER) CYCLE					
Lung (LU)	Shaoshang LU11	Yuji LU10	Taiyuan LU9	Jingqu LU8	Chize LU5
Large Intestine (LI)	Shangyang LI1	Erjian LI2	Sanjian LI3	Yangxi LI5	Quchi LI11
Stomach (ST)	Lidui ST45	Neiting ST44	Xiangyu ST43	Jiexi ST41	Zusanli ST36
Spleen (SP)	Yinbai SP1	Dadu SP2	Taibai SP3	Shangqiu SP5	Yinlingquan SP9
POSTERIOR (INNER) CYCLE					
Heart (HT)	Shaochong HT9	Shaofu HT8	Shenmen HT7	Lingdao HT4	Shaohai HT3
Small Intestine (SI)	Shaoze SI1	Qiangyu SI2	Houxi SI3	Yanggu SI5	Xiaohai SI8
Bladder (BL)	Zhiyin BL67	Zutonggu BL66	Shugu BL65	Kunlun BL60	Weizhong BL40
Kidney (KI)	Yongquan KI1	Rangu KI2	Taixi KI3	Fuliu KI7	Yingu KI10
MIDDLE (LATERAL) CYCLE					
Pericardium (PC)	Zhongchong PC9	Laogong PC8	Daling PC7	Jianshi PC5	Quze PC3
Triple Energizer (SJ/TE)	Guanchong SJ1	Yemen SJ2	Zhongzhu SJ3	Zhigou SJ6	Tianjing SJ10
Gallbladder (GB)	Zuqiaoyin GB44	Xiaxi GB43	Zulinqi GB41	Yangfu GB38	Yanglingquan GB34
Liver (LR)	Dadun LR1	Xingjian LR2	Taichong LR3	Zhongfeng LR4	Ququan LR8

Yin meridians have no separate Yuan-Source point -- the Shu-Stream point IS the Yuan point (e.g., Taiyuan LU9 is both Shu-Stream and Yuan for the Lung). Yang meridians have a 6th, separate Yuan-Source point beyond the 5 Shu points (e.g., Chongyang ST42 for the Stomach).

The Eight Confluent Points

Connect the 8 Extraordinary Vessels to the 12 Regular Meridians

Confluent points are special acupuncture points that connect the Eight Extraordinary Vessels with the Twelve Regular Meridians. Function: helping to regulate the body's energy, connect related meridians, and treat specific patterns.

Houxi (SI 3)

Du Vessel · pairs w/ Shenmai (BL 62)

On the ulnar side of the hand, proximal to the 5th metacarpophalangeal joint, at the border of the red and white skin.

Benefits the spine and neck; clears heat; treats febrile disease and back pain.

Lieque (LU 7)

Ren Vessel · pairs w/ Zhaohai (KI 6)

On the radial forearm, 1.5 cun proximal to the wrist crease, superior to the styloid process of the radius.

Releases the exterior, benefits the throat and lungs, and regulates the Ren Vessel.

Gongsun (SP 4)

Chong Vessel · pairs w/ Neiguan (PC 6)

On the medial foot, distal and inferior to the base of the 1st metatarsal bone.

Harmonizes the middle jiao, regulates the Chong Vessel, and treats abdominal or menstrual disorders.

Zulinqi (GB 41)

Dai Vessel · pairs w/ Waiguan (SJ 5)

On the dorsum of the foot, distal to the junction of the 4th and 5th metatarsal bones.

Spreads Liver qi, regulates the Dai Vessel, benefits the breasts, and treats pelvic or lateral-body pain.

Zhaohai (KI 6)

Yin Qiao Vessel · pairs w/ Lieque (LU 7)

In the depression directly below the medial malleolus.

Nourishes yin, benefits the throat, regulates sleep; governs movement.

Shenmai (BL 62)

Yang Qiao Vessel · pairs w/ Houxi (SI 3)

In the depression directly below the lateral malleolus.

Regulates the Yang Qiao Vessel, benefits the eyes, calms the spirit, and governs motor function.

Neiguan (PC 6)

Yin Wei Vessel · pairs w/ Gongsun (SP 4)

2 cun proximal to the wrist crease, between the tendons of palmaris longus and flexor carpi radialis.

Opens the chest, regulates the Heart, calms the spirit, harmonizes the Stomach, and relieves nausea.

Waiguan (SJ 5)

Yang Wei Vessel · pairs w/ Zulinqi (GB 41)

2 cun proximal to the dorsal wrist crease, between the radius and ulna.

Releases the exterior, clears heat, benefits the head and ears, and relieves pain along the yang channels.

The 8 points form 4 master couples (each couple treats a shared body region): Houxi+Shenmai (inner canthus, ear, shoulder, neck -- Yang); Lieque+Zhaohai (lung, throat, chest -- Yin); Gongsun+Neiguan (heart, chest, stomach -- Yin); Zulinqi+Waiguan (outer canthus, ear, cheek, neck, shoulder -- Yang).

The 15 Collaterals — Master Table

Full detail + clinical indications: Week 8 Decoder

The twelve Primary Meridians and the Conception and Governor Vessels each give off a collateral branch, totaling fifteen when combined with the major collateral of the Spleen -- collectively known as the fifteen collaterals. These are named after the acupoints from which they originate.

The twelve collateral channels branch out from the Luo-Connecting points of their respective meridians and travel toward their internally-externally related paired meridians. The major collateral channels of the Ren, Du, and Spleen meridians branch out from their own Luo-Connecting points, spreading across the abdomen, back of the head, and chest-rib regions.

MERIDIAN	LUO POINT	CONNECTS TO
Lung (LU)	Lieque LU7	Large Intestine
Large Intestine (LI)	Pianli LI6	Lung
Stomach (ST)	Fenglong ST40	Spleen
Spleen (SP)	Gongsun SP4	Stomach
Heart (HT)	Tongli HT5	Small Intestine
Small Intestine (SI)	Zhizheng SI7	Heart
Bladder (BL)	Feiyang BL58	Kidney
Kidney (KI)	Dazhong KI4	Bladder
Pericardium (PC)	Neiguan PC6	Triple Energizer
Triple Energizer (SJ)	Waiguan SJ5	Pericardium
Gallbladder (GB)	Guangming GB37	Liver
Liver (LR)	Ligou LR5	Gallbladder
Conception Vessel (Ren)		Jiuwei CV15
Governor Vessel (Du)		Changqiang GV1
Major Collateral of the Spleen		Dabao SP21

The 3 Extra Collaterals — running course:

Conception Vessel (Ren) (Jiuwei CV15): Separates from the Governor Vessel at the lower end of the sternum; from CV15 it spreads over the abdomen.

Governor Vessel (Du) (Changqiang GV1): Arises in the perineum, runs upward along both sides of the spine to the nape, spreads over the top of the head; at the scapular region it connects with the Bladder meridian and pierces through the spine.

Major Collateral of the Spleen (Dabao SP21): Begins from Dabao SP21, emerges 3 cun below Yuanye GB22, and spreads through the chest and hypochondriac region, gathering blood all over the body.

Middle Circuit Divergent Channels (PC · SJ · GB · LR):

Each is described by Li (beginning) / He (organs involved) / Chu (exiting) / Ru (merging to). Divergent Channels have NO acupoints of their own.

MERIDIAN	BEGINNING (LI)	ORGANS/SYSTEMS (HE)	EXITING (CHU) / MERGING (RU)
Pericardium (PC)	Chest / 3 cun below the axilla	Sanjiao, throat	Behind the ear -> Sanjiao meridian (Hand-Shaoyang)
Triple Energizer (SJ)	Vertex / supraclavicular fossa	Chest, upper/middle/lower jiao	Supraclavicular fossa -> Rejoins its own primary meridian
Gallbladder (GB)	Lateral thigh / lower abdomen (pelvic region)	Gallbladder, Liver, Heart, esophagus, eye	Face / outer canthus -> Gallbladder meridian of Foot-Shaoyang
Liver (LR)	Instep of the foot / pelvic region	Pubic region	Pubic region -> Converges with the Gallbladder divergent meridian

PC + SJ divergent channels converge with each other; GB + LR divergent channels converge with each other -- Divergent Channels always travel in Yin/Yang paired-meridian sets before merging into the Yang meridian.

Middle Circuit — Muscle (Sinew) Region Pathways

PC · SJ · GB · LR

General Functions of the 12 Muscle Regions:

- Connects all the bones and joints of the body and maintains normal range of motion.
- Originates from the extremities of the limbs and ascends to the head and trunk, but does not reach the internal organs.
- Clinical significance: muscular problems -- Bi syndrome, contracture, stiffness, spasm, muscular atrophy.

Hand Jueyin (Pericardium)

Middle finger -> elbow -> axilla -> ribs -> chest -> axilla -> spreads over the chest -> thoracic diaphragm.

Hand Shaoyang (Sanjiao)

4th finger -> wrist -> forearm -> olecranon -> upper arm -> shoulder -> neck -> joins Hand-Taiyang sinew -> angle of mandible -> root of tongue -> in front of the ear -> outer canthus -> temple -> corner of the forehead.

Foot Shaoyang (Gallbladder)

4th toe -> external malleolus -> tibia -> knee -> fibula -> thigh (Futu ST32) -> sacrum. Straight branch: ribs -> axilla -> breast -> Quepen ST12 -> behind the ear -> temple -> vertex. Branch: temple -> cheek -> bridge of nose -> outer canthus.

Foot Jueyin (Liver)

Dorsum of the great toe -> medial malleolus -> tibia -> knee -> thigh -> genital region -> joins the other Muscle Regions.

Hand Jueyin: Medial side of elbow, axilla, diaphragm | Hand Shaoyang: Dorsum of wrist, posterior elbow, corner of forehead | Foot Shaoyang: Outer canthus & side of nose, SC fossa (ST12), sacrum, above ST32, lateral malleolus & lateral knee | Foot Jueyin: Medial malleolus (anterior), tibia (medial condyle), genitals

The 4 Structural Pattern Rules (all 12 Muscle Regions):

- All 3 Yang Muscle Regions of the foot connect with the eyes.
- All 3 Yin Muscle Regions of the foot connect with the genital region.
- All 3 Yang Muscle Regions of the hand connect with the angle of the forehead.
- All 3 Yin Muscle Regions of the hand connect with the thoracic cavity.

Middle Circuit — Muscle Regions, Lecture Figures

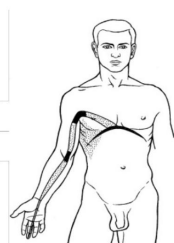
PC · SJ · GB · LR

Hand Jueyin (Pericardium)

Muscle Region of Hand Jueyin (Pericardium)

➤ **Muscle Region of Hand Jueyin (Pericardium)**

Middle finger ---> elbow ---> axilla ---> ribs ---> chest --->
axilla ---> spreads over the chest ---> thoracic diaphragm.



Pericardium

Binds (i)
Medial side of elbow
Axilla
Diaphragm

20

Binds: Medial side of elbow, axilla, diaphragm

Hand Shaoyang (Sanjiao)

Muscle Region of Hand Shaoyang (Sanjiao)

➤ **Muscle Region of Hand Shaoyang (Sanjiao)**

Fourth finger ---> wrist ---> forearm ---> olecranon of the elbow -
---> upper arm ---> shoulder ---> neck ---> Muscle Region of the
Hand Taiyang ---> angle of the mandible ---> root of the tongue ---
> in front of the ear ---> outer canthus ---> temple ---> corner of
the forehead.



SanJiao

Binds (i)
Dorsum of wrist
Posterior part of elbow
Corner of forehead

21

Binds: Dorsum of wrist, posterior elbow, corner of forehead

Foot Shaoyang (Gallbladder)

Muscle Region of Foot Shaoyang (Gallbladder)

➤ **Muscle Region of Foot Shaoyang (Gallbladder)**

The fourth toe ---> external malleolus ---> tibia ---> knee. Fibula --->
thigh (Futu, St 32). ---> sacrum.
Straight branch ---> ribs ---> axilla ---> breast region ---> Quepen (St
12). ---> behind the ear ---> temple ---> vertex.
Branch: temple ---> cheek ---> bridge of the nose ---> outer canthus.



Gallbladder

Binds (i)
Outer canthus & side of nose
SC fossa (ST12) region
Sacrum
Just superior to ST32
Lateral malleolus, lateral aspect of knee

28

Binds: Outer canthus & side of nose, SC fossa (ST12), sacrum, above ST32, lateral malleolus & lateral knee

Foot Jueyin (Liver)

Muscle Region of Foot Yueyin (Liver)

➤ **Muscle Region of Foot Yueyin (Liver)**

Dorsum of the big toe ---> medial malleolus ---> tibia --->
knee ---> thigh ---> genital region ---> other Muscle Regions.



Liver

Binds (i)
Medial malleolus (anterior)
Tibia (medial condyle)
Genitals

29

Binds: Medial malleolus (anterior), tibia (medial condyle), genitals

Lecture Fig. — 2026AC300Lecture_9Vivian.pdf, slides 20/21/28/29 (Dr. Zhang).

The 12 Cutaneous Regions (Pi Bu)

Fully new content this week

The twelve cutaneous parts are the parts of the twelve meridians reflected on the body surface, where the Qi of the meridians is distributed -- the outermost layer of the human body.

Su Wen (Plain Questions, Ch. 56): "The Cutaneous Regions are the part of the meridian system located in the superficial layers of the body. The Cutaneous Regions are marked by the regular meridians." Transmitting order of a disease: "Skin -> Collaterals -> Meridians -> Fu organs -> Zang organs." Diagnostic color rule: "Blue-colored skin signifies local pain. Dark-colored skin indicates blockage of qi and blood. Yellow to red colored skin refers to heat syndromes, and white colored skin to cold syndromes."

Functions:

- Protects the organism from exogenous pathogen invasion -- the first line of defense.
- Projects symptoms and signs of internal disease onto the body surface (diagnostic skin-color rule).
- Interacts diagnostically and therapeutically -- the theoretical basis for gua sha, cupping, and intradermal/press needles.

Exception to the normal rule: Cutaneous Regions are the one structure that connects the Hand and Foot meridians of the SAME name together (e.g., Taiyang of the hand + Taiyang of the foot form one continuous Taiyang cutaneous region) -- this is why there are only 6 cutaneous regions, not 12.

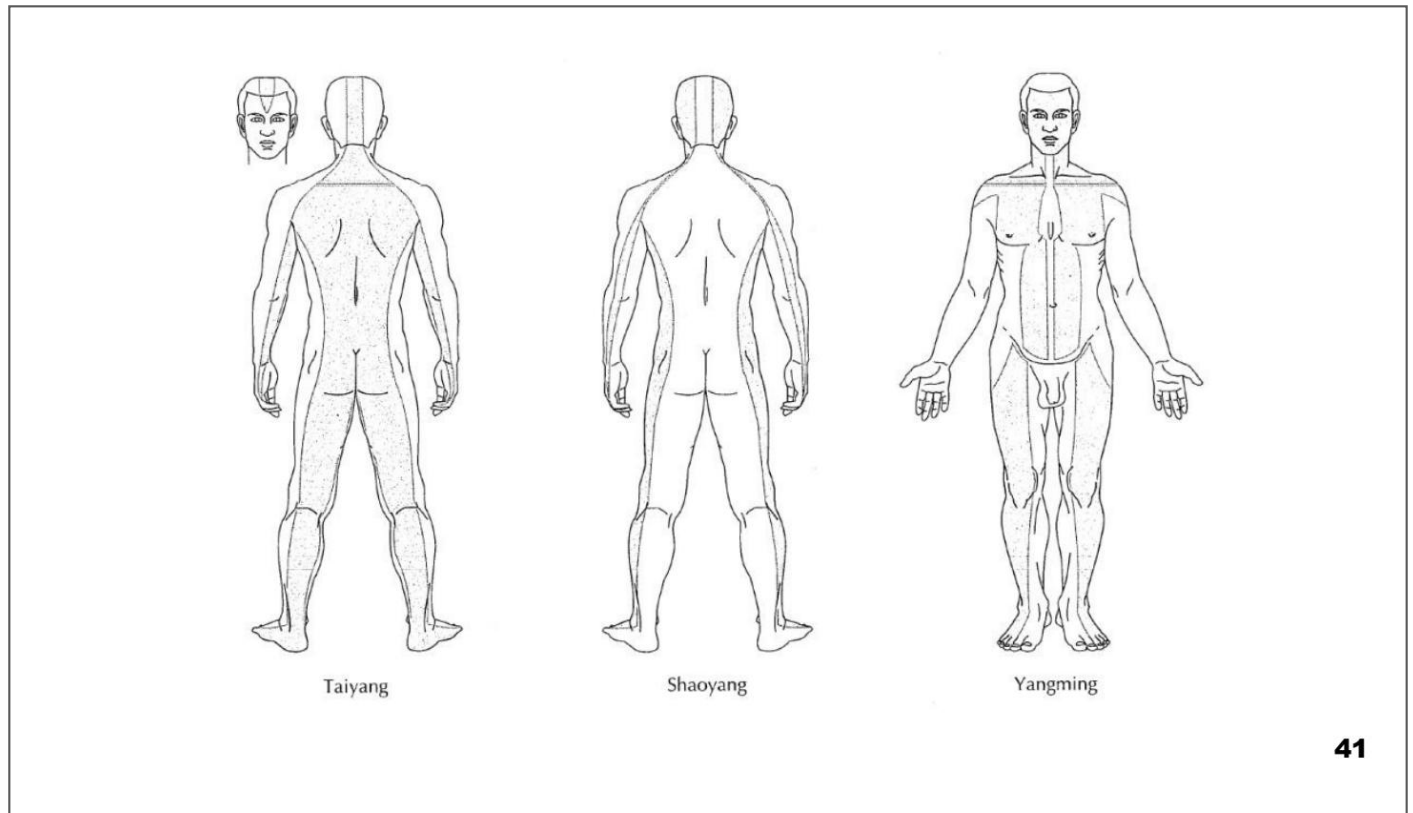
The 6 Cutaneous Groups:

Taiyang	BL (foot) · SI (hand)
Shaoyang	GB (foot) · SJ (hand)
Yangming	ST (foot) · LI (hand)
Taiyin	SP (foot) · LU (hand)
Shaoyin	KI (foot) · HT (hand)
Jueyin	LR (foot) · PC (hand)

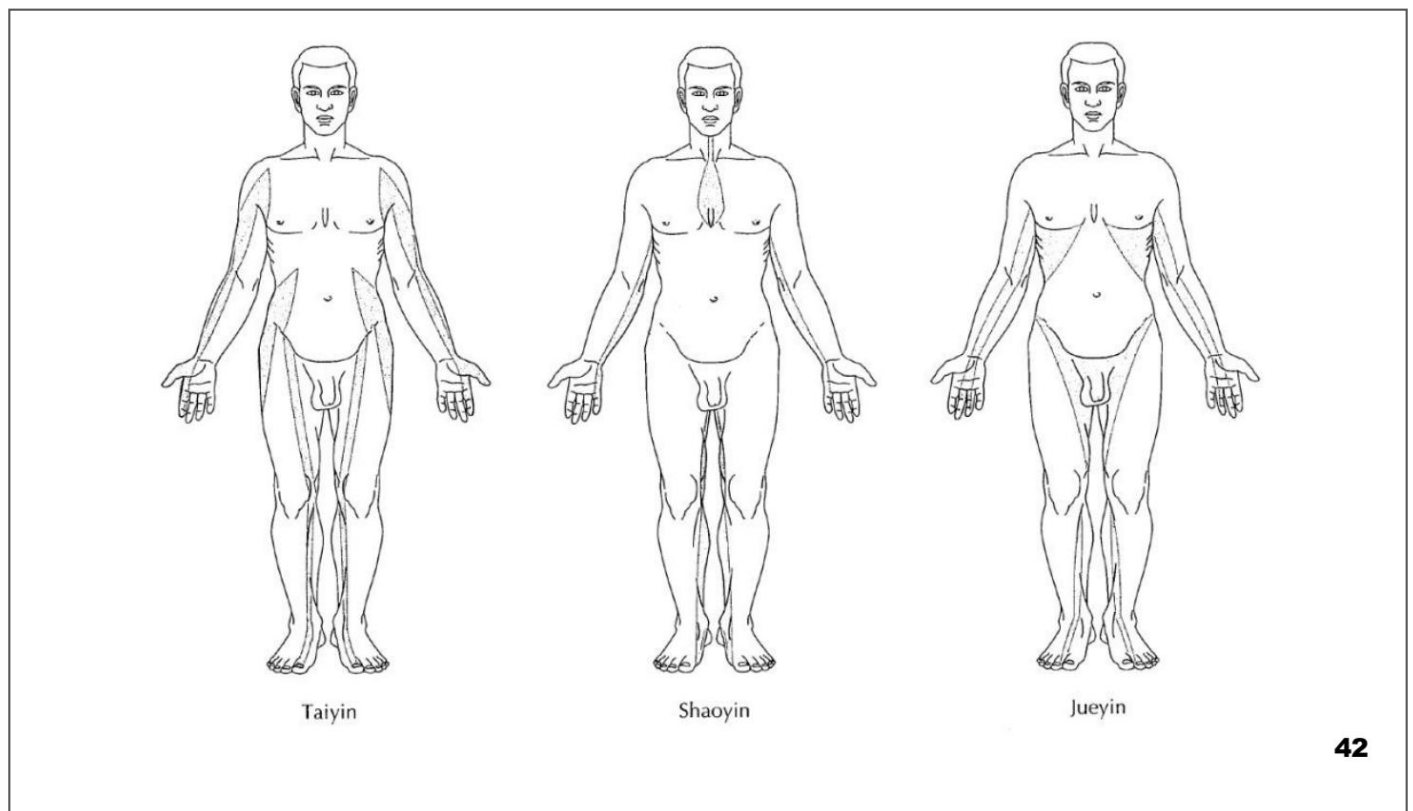
12 Cutaneous Regions — Lecture Figures

Yang groups (top) · Yin groups (bottom)

Yang Groups: Taiyang · Shaoyang · Yangming



Yin Groups: Taiyin · Shaoyin · Jueyin



Lecture Fig. — 2026AC300Lecture_9Vivian.pdf, slides 41-42 (Dr. Zhang).

Final Exam Master Review — Circulation & Structure

The single highest-yield page in this packet

Dr. Zhang's own exam-focus summary (slide 57, 2026 deck): "The exam focuses on the pathways, connections, distribution, and special features of the meridians. You should know which organs each meridian is associated with, their starting and ending points, and key landmarks they pass -- such as the ear, nose, inner canthus, external genitalia, pubic region, and waist. Pay attention to the unique traits of certain meridians, the running course of the channel, the direction of Qi flow. Also review paired relationships between meridians, the composition of the fifteen collaterals, and how channels connect with each other." Keyword list from the same slide: 12 primary Meridians, Belt Vessel, GV, 15 Collaterals.

The 12 Meridians Circulate in a Fixed Sequence:

1. 3 Yin meridians of the hand run chest -> hand, converging with the 3 Yang meridians of the hand (exterior-interior pairs) at the fingertips.
2. 3 Yang meridians of the hand ascend from the fingertips to the head, where they connect with the 3 Yang meridians of the foot.
3. 3 Yang meridians of the foot descend from the head to the toes, where they join the 3 Yin meridians of the foot.
4. 3 Yin meridians of the foot ascend from the toes to the abdomen and chest, meeting the 3 Yin meridians of the hand -- closing the full circuit.

Direction of Qi Flow:

Yin meridians of the Hand	Chest -> Hand
Yang meridians of the Hand	Hand -> Head
Yang meridians of the Foot	Head -> Foot
Yin meridians of the Foot	Foot -> Abdomen/Chest

Distribution Rules:

Limbs -- medial aspect:

Yin meridians: Anterior=Taiyin, Middle=Jueyin, Posterior=Shaoyin

Limbs -- lateral aspect:

Yang meridians: Anterior=Yangming, Middle=Shaoyang, Posterior=Taiyang

Head & Trunk:

Anterior=Yangming, Posterior=Taiyang, Lateral=Shaoyang

Which Meridians Pass the Eye, and Where:

Bladder (BL)	Starts at the inner canthus (BL1 Jingming).
Gallbladder (GB)	Starts at the outer canthus (GB1 Tongziliao).
Small Intestine (SI)	A branch goes to the outer canthus; another travels below the eye to the inner canthus (only meridian connecting BOTH).
Sanjiao (SJ)	A branch runs from behind the ear to the outer canthus.
Liver (LR)	Follows the throat upward, connects to the 'Eye System.'
Heart (HT)	A branch links the heart to the 'Eye System' via the throat.
Stomach (ST)	Ascends to the bridge of the nose, where it meets the Bladder meridian at the inner canthus.

The 8 Extraordinary Vessels — Key Structural Differences:

The 8 Extraordinary Vessels have NO internal Zang-Fu connections (no pertaining/connecting organ) -- their only running course is superficial (nape, vertex, hip, etc.), which is precisely what makes them different from the 12 Primary Meridians. Du, Ren, and Chong all begin in the lower abdomen -- 'one source, three branches.' The Belt (Dai) Vessel is the only vessel that runs horizontally around the waist rather than up/down the body. Yin Qiao and Yang Qiao both begin at their respective Confluent points (KI6 / BL62).

Final Exam Master Table — All 12 Primary Meridians

Lung (LU)

Total pts: 11 · Chest -> Hand

Pertains: Lung | Connects: Large Intestine
 First: Zhongfu LU1 (chest) | Last: Shaoshang LU11 (thumb, radial)
Passes through the medial anterior border of the upper limb.

Large Intestine (LI)

Total pts: 20 · Hand -> Head

Pertains: Large Intestine | Connects: Lung
 First: Shangyang LI1 (index finger) | Last: Yingxiang LI20 (nasolabial groove)
Ends beside the nose, linking to the Stomach meridian.

Stomach (ST)

Total pts: 45 · Head -> Foot

Pertains: Stomach | Connects: Spleen
 First: Chengqi ST1 (below pupil) | Last: Lidui ST45 (2nd toe)
Starts lateral to the ala nasi; links to Spleen at the great toe (SP1).

Spleen (SP)

Total pts: 21 · Foot -> Chest/Abdomen

Pertains: Spleen | Connects: Stomach
 First: Yinbai SP1 (great toe) | Last: Dabao SP21 (lateral chest, Major Luo)
Ascends alongside the esophagus, reaches the tongue (connects to Heart via internal branch).

Heart (HT)

Total pts: 9 · Chest -> Hand

Pertains: Heart | Connects: Small Intestine
 First: Jiquan HT1 (axilla) | Last: Shaochong HT9 (little finger, radial)
Connects to the 'Eye System' via the throat.

Small Intestine (SI)

Total pts: 19 · Hand -> Head

Pertains: Small Intestine | Connects: Heart
 First: Shaoze SI1 (little finger, ulnar) | Last: Tinggong SI19 (anterior to tragus)
Only meridian connecting BOTH inner and outer canthus of the eye.

Bladder (BL)

Total pts: 67 · Head -> Foot

Pertains: Bladder | Connects: Kidney
 First: Jingming BL1 (inner canthus) | Last: Zhiyin BL67 (small toe, lateral)
Largest meridian in the body (67 points, 4 branches); starts at inner canthus.

Kidney (KI)

Total pts: 27 · Foot -> Chest

Pertains: Kidney | Connects: Bladder
 First: Yongquan KI1 (sole) | Last: Shufu KI27 (below clavicle)
Connecting organs also include Liver and Lung; links with Pericardium at the chest.

Pericardium (PC)

Total pts: 9 · Chest -> Hand

Pertains: Pericardium | Connects: Triple Energizer (Sanjiao)
 First: Tianchi PC1 (4th ICS) | Last: Zhongchong PC9 (middle finger tip)
Links with Sanjiao at the ring finger via a branch from Laogong PC8.

Triple Energizer (SJ/TE)

Total pts: 23 · Hand -> Head

Pertains: Sanjiao (Upper/Middle/Lower Jiao) | Connects: Pericardium
 First: Guanchong SJ1 (ring finger) | Last: Sizhukong SJ23 (lateral eyebrow)
Runs in front of the ear; terminates at the outer canthus, links to Gallbladder.

Gallbladder (GB)

Total pts: 44 · Head -> Foot

Pertains: Gallbladder | Connects: Liver
 First: Tongziliao GB1 (outer canthus) | Last: Zuqiaoyin GB44 (4th toe)
Zig-zag pathway across the head; only meridian to trace this shape.

Liver (LR)

Total pts: 14 · Foot -> Chest

Pertains: Liver | Connects: Gallbladder
 First: Dadun LR1 (great toe, dorsal hairy region) | Last: Qimen LR14 (Front-Mu of Liver, 6th ICS)
Passes through the external genitalia and the 'Eye System'; links to Lung at the chest, closing the 12-meridian cycle.

Clinical Evidence — PCOS/PMOS, Incontinence & EMMA

Appendix -- not exam material

PCOS/PMOS Acupuncture Protocol

Polycystic Ovary Syndrome (PCOS) is the most common cause of anovulatory infertility, affecting an estimated 170 million women worldwide. In 2026, PCOS was renamed PMOS (Polyendocrine Metabolic Ovarian Syndrome), highlighting the central role of metabolic dysfunction -- particularly insulin resistance (IR) -- rather than treating it as a purely reproductive/ovarian condition. It is a common metabolic AND reproductive disorder: menstrual irregularity, hirsutism, sleep disruption, and a strong tendency toward obesity. No single first-line treatment currently addresses PCOS/PMOS's multisystem dysfunction, and adherence to first-line treatment is poor -- motivating an AI-enabled, acupuncture-based intervention targeting the metabolic mechanism directly.

Study Design:

International multicenter RCT + case-control collaboration (first industry-standard registration in this area; protocol validated through 3 Delphi rounds, adopted by the Karolinska Institutet research team).

Dosing:

Manual acupuncture combined with electroacupuncture, 5 sessions/week, 30 min/session (2-3x/week is also effective per Dr. Zhang's clinical experience); 3-4 months total treatment, since one menstrual cycle = one treatment unit for reproductive/gynecological cases.

Protocol 1:

Protocol 1: CV4-CV6, ST29-ST29, ST32-ST34 (bilateral). Needles placed in muscle (15-35mm), manually stimulated to de qi. Electrical stimulation as high as tolerable without pain (Program 10). Needle gauge scaled to patient BMI (0.25x30mm low BMI -> 0.30x50/75mm obese).

Protocol 2 (alternated):

Protocol 2: CV6-CV10, ST27-ST27, SP10 + an extra point 6 cun cranial in vastus lateralis. Same stimulation parameters as Protocol 1. The two protocols are ALTERNATED across the treatment course ("Phlegm-Dampness Obesity" framework); Sanyinjiao/Xuehai (activate blood, resolve stasis), Zhongwan/Zusanli/Fenglong (strengthen Spleen, drain dampness).

Outcomes:

BMI: 30.18 -> 26.57 (p<0.01). Insulin resistance (HOMA-IR): 5.26 -> 2.89 (p<0.05). Hyperandrogenism (SHBG/FAI): 12.76 -> 7.46 (p<0.01). No rebound at 4-month follow-up (effect held with zero further treatment during the follow-up window).

Basic Reference Set:

Basic reference acupuncture protocol for infertility/gynecology cases (Dr. Zhang's own clinical framework): GV20, CV20, CV12, CV6, GV4, PC6, ST16, ST6, SP6. Can be expanded or reduced by symptom and patient presentation; typical dosing 2x/week for 3 weeks.

TEACHING ACUPOINT SET -- NOT A PRESCRIPTION. Acupoints may support comfort, but no point set is proven to improve IVF live-birth rates. ST36/SP6 support general constitutional/menstrual symptoms; PC6 supports nausea/anxiety-related symptoms; CV3/CV4/CV6 are lower-abdominal points used in some fertility protocols. Classroom orientation only -- point selection, depth, stimulation, and timing require individualized clinical assessment.

Urinary Incontinence & EMMA Pilot Data

Stress incontinence: Leakage with coughing, sneezing, running, jumping, or lifting.
Urgency incontinence: A sudden compelling urge followed by leakage; may include frequency and nocturia.
Mixed incontinence: Stress and urgency symptoms occur together.

JAMA 2017 Electroacupuncture RCT (Liu Z, et al. JAMA. 2017;317(24):2493-2501. doi:10.1001/jama.2017.7220):

Multicenter RCT, 12 hospitals, 504 women, ages 40-75. Electroacupuncture vs. sham electroacupuncture (blunt placebo needles, no penetration/current/de qi).

Bilateral Zhongliao (BL33) + Huiyang (BL35), lumbosacral electroacupuncture. 0.30 x 75mm needles, depth approx. 50-60mm. BL33 inserted 30°-45° inferomedially; BL35 slightly superolaterally. Twirl/lift-thrust to de qi, then attach paired electrodes. Continuous wave, 50Hz, 1-5mA, 30 min/session.

3 sessions/week (ideally every other day) x 6 consecutive weeks = 18 sessions total.

Primary outcome: change in urine leakage at week 6, measured by the standardized 1-hour pad test.

Research points, not a substitute for pelvic-floor training. NICE does not recommend complementary therapies routinely for urinary incontinence or overactive bladder. Avoid home needling or direct moxibustion at the sacrum -- burns, smoke exposure, and delayed diagnosis are real risks.

Prompt assessment needed for: blood in urine, pain/fever, recurrent urinary infection, voiding difficulty or retention, neurological symptoms, a vaginal bulge, or sudden marked worsening.

EMMA Robotic Massage — Pelvic Pain Pilot Data:

EMMA (Expert Manipulative Massage Automation) is a robot designed to deliver Tuina-style manipulative massage, developed in collaboration with a Singapore-based team; Dr. Zhang uses a version of this technology in her own clinic (gynecology focus: qi/pelvic pain).

22 registered records; 21 participants treated. Course maturity at data cut (15 Aug 2026): 22 registered -> 21 started -> 18 reached >=3 sessions -> 13 reached >=5 sessions -> 10 completed all 6 sessions. Data instruments: 96 paired pre/post VAS records; NRS, McGill Pain Questionnaire, SF-36, and mood measures at checkpoints; 126 thermography file/range references linked to pain phenotype, menstrual context, and height/weight.

Fixed 6-session completer cohort (n=10, 60 paired treatment visits). Mean VAS fell from 6.2 pre-treatment (Session 1) to 1.8 post-treatment (Session 6) -- a 71.9% reduction (6.22 -> 1.75 course average, -4.47 points). 57/60 visits showed a lower post-session VAS than pre-session. 10/10 participants (100%) achieved >=30% reduction; 9/10 (90%) achieved >=50% reduction.

The repeated within-patient signal is strong enough to justify prospective controlled validation with predefined responder endpoints -- translating established manual treatment approaches into robotic interventions, not replacing the clinician's assessment and point selection.